Inverse kinematics

Given a desired configuration for the tool frame, find the joint angles that will achieve this configuration.

Given forward kinematics map $g_{st} : Q \to SE(3)$ and a desired configuration $g_d \in SE(3)$, solve

$$g_{st}(\theta) = g_d$$

for some $\theta \in Q$.

Multiple solutions or a unique solution or no solution.

Can be a difficult problem to solve.

§3.2 Paden-Kahan subproblems

- Based on the product of exponentials formula
- First build a library of subproblems that occur frequently, with solutions
- Once a library is available, one seeks to reduce the full inverse kinematics problem into appropriate subproblems
- 3 subproblems are derived in MLS §3.2. These already allows the solution of the inverse kinematics problem for a significant number of manipulators.
- More subproblems are stated in exercises.

§3.2 Paden-Kahan subproblems

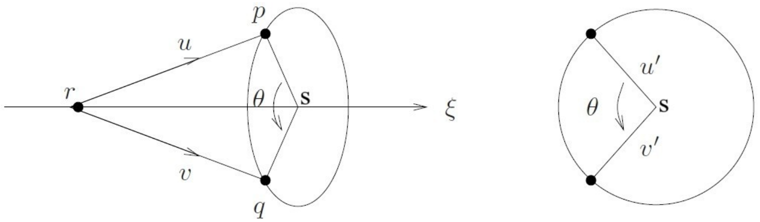
Subproblem 1: Rotation about a single axis
Let ξ be a zero-pitch twist with unit magnitude and $p, q \in \mathbb{R}^3$ two points. Find θ such that $e^{\hat{\xi}\theta} p = q$.

Subproblem 2: Rotation about two subsequent axes
Let ξ_1 and ξ_2 be two zero-pitch, unit magnitude twists with intersecting axes and $p, q \in \mathbb{R}^3$ two points. Find θ_1 and θ_2 such that $e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} p = q$.

Subproblem 3: Rotation to a given distance
Let ξ be a zero-pitch, unit magnitude twist; $p, q \in \mathbb{R}^3$ two points; and $\delta \in \mathbb{R}; \delta > 0$. Find θ such that $||q - e^{\hat{\xi}\theta} p|| = \delta$.

Subproblem 1: Rotation about a single axis

Let ξ be a zero-pitch twist with unit magnitude and $p, q \in \mathbb{R}^3$ two points. Find θ such that $e^{\hat{\xi}\theta} p = q$.



$$u = p - r \qquad\qquad v = q - r$$

Since $e^{\hat{\xi}\theta} r = r$ (note implied vector dimension differences):

$$e^{\hat{\xi}\theta}(u + r) = v + r \quad \implies \quad e^{\hat{\omega}\theta} u = v \quad (3.18)$$

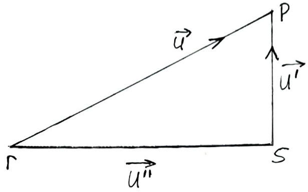
Subproblem 1: Rotation about a single axis

$$\vec{u} = \vec{u''} + \vec{u'}$$

$$u = \omega\omega^T u + u'$$

$$u' = u - \omega\omega^T u$$

$$v' = v - \omega\omega^T v$$



Solution if

$\omega^T u = \omega^T v$ & $||u'|| = ||v'||$

Subproblem 1: Rotation about a single axis

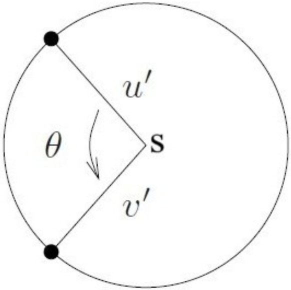
$$\widehat{u'}v' = ||u'||\;||v'||\omega \sin\theta$$

$$\omega^T\;\widehat{u'}v' = ||u'||\;||v'||\sin\theta$$

$$(u')^T\;v' = ||u'||\;||v'||\cos\theta$$

$$\tan\theta = \frac{\omega^T\;\widehat{u'}v'}{(u')^T\;v'}$$

$$\theta = \operatorname{atan2}\left(\omega^T\;\widehat{u'}v', (u')^T\;v'\right)$$



Subproblem 2: Rotation about a single axis

Explained in class using textbook.

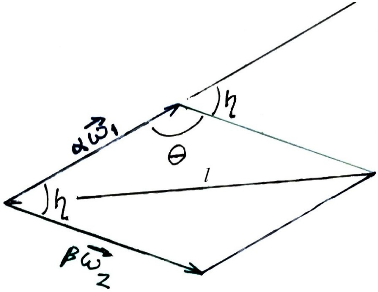
For MLS eq. (3.25) p. 102: The vector sum $\alpha\vec{\omega}_1 + \beta\vec{\omega}_2$ has a length l calculated with the cosine rule ($||\omega_1|| = ||\omega_2|| = 1$):

$$
\begin{aligned}
l^2 &= \alpha^2 + \beta^2 - 2\alpha\beta\cos\theta \\
&= \alpha^2 + \beta^2 + 2\alpha\beta\cos\eta
\end{aligned}
$$

But $\cos\eta = \omega_1^T\omega_2$.

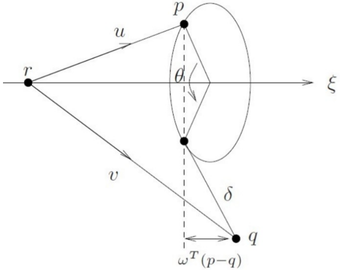
Therefore

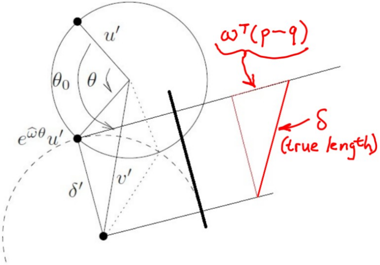
$$l^2 = \alpha^2 + \beta^2 + 2\alpha\beta\omega_1^T\omega_2$$



Subproblem 3: Rotation to a given distance

Explained in class using textbook.





$$\delta'^2 + ||\omega^T(p-q)||^2 = \delta^2 \quad \Longrightarrow \quad \delta'^2 = \delta^2 - ||\omega^T(p-q)||^2$$

$$||v' - e^{\hat{\omega}\theta}u'||^2 = \delta'^2$$

§3.3 Solving inverse kinematics using subproblems

3rd example (unnumbered in text above Ex 3.5):

$$
\begin{aligned}
\delta := ||gp - q|| &= ||e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} e^{\hat{\xi}_3\theta_3} p - q|| \\
&= ||e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} e^{\hat{\xi}_3\theta_3} p - e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} q|| \\
&= ||e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} \left(e^{\hat{\xi}_3\theta_3} p - q \right) || \\
&= ||e^{\hat{\xi}_3\theta_3} p - q||
\end{aligned}
$$